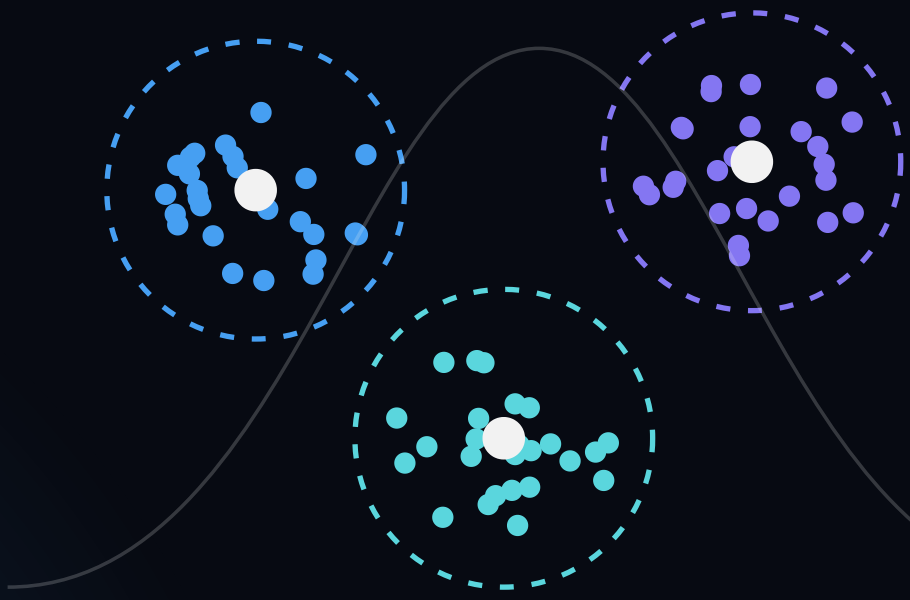
NeuralVarsity.ai

PROGRAM 02 · 6 MONTHS

Data Science with GenAI

Master the full data-science lifecycle — and ship it with GenAI.



6 Months	120	13	34
24 WEEKS	SESSIONS	PROJECTS	TOOLS

PROGRAM OVERVIEW

The complete data-science lifecycle — from Python to production GenAI.

A 24-week, 120-session program that takes you from Python and statistics through machine learning, deep learning and computer vision, NLP and Generative AI, to SQL and Power BI — finishing with an end-to-end analytics system.

Who it's for

- Aspiring data scientists who want the complete lifecycle
- Analysts ready to move into machine learning
- Engineers and developers transitioning into AI
- STEM graduates building a data-science portfolio

At a glance

- **Level:** Beginner to advanced
- **Starting point:** Starts from programming fundamentals and environment setup.
- **Format:** Day-by-day, hands-on
- **Generative AI:** OpenAI, DALL·E, Gemini, LLaMA, LangChain RAG and a deployed GenAI assistant

What you'll be able to do

- Write clean, object-oriented Python and wrangle data with NumPy and Pandas
- Run rigorous EDA, probability and hypothesis testing — A/B, chi-square, ANOVA
- Train and tune supervised and unsupervised models, from regression to XGBoost, K-Means and DBSCAN
- Build deep networks with TensorFlow and Keras and detect objects in real time with YOLO
- Process language with NLTK, spaCy, embeddings, RNNs, LSTMs, Transformers and BERT
- Build GenAI apps with OpenAI, Gemini, LLaMA and LangChain RAG, deployed with Streamlit
- Query databases with advanced SQL and tell the story in Power BI

Roles this program prepares you for

Data Scientist | Machine Learning Engineer | GenAI Developer | Computer Vision Engineer | NLP Engineer | Data / BI Analyst

PROGRAM STRUCTURE

24 weeks · 6 months · 120 sessions

Month 01	**Python Programming, Data Handling & EDA Foundation**	Weeks 1–4 · Days 1–20 · 20 sessions
Month 02	**Statistics, Exploratory Data Analysis & Probability**	Weeks 5–8 · Days 21–40 · 20 sessions
Month 03	**Supervised & Unsupervised Machine Learning**	Weeks 9–12 · Days 41–60 · 20 sessions
Month 04	**Deep Learning & Computer Vision**	Weeks 13–16 · Days 61–80 · 20 sessions
Month 05	**Natural Language Processing (NLP) & Generative AI**	Weeks 17–20 · Days 81–100 · 20 sessions
Month 06	**SQL Complete & Business Intelligence (Power BI)**	Weeks 21–24 · Days 101–120 · 20 sessions

Tools & technologies

Python · Anaconda · Jupyter · VS Code · Google Colab · NumPy · Pandas · Matplotlib · Seaborn · Regex · scikit-learn · XGBoost · TensorFlow · Keras · TensorBoard · OpenCV · YOLO (Ultralytics) · NLTK · spaCy · BERT · OpenAI API · DALL·E · Gemini · Meta LLaMA · LangChain · Streamlit · Git · GitHub · MySQL Workbench · SQL · Excel · Power BI · Power Query · DAX

Month 01 — Python Programming, Data Handling & EDA Foundation

Weeks 1–4 · Days 1–20

WEEK 1 · Core Programming and Data Structures

Day 1 — Introduction to Programming & Environment Setup

A foundational start, defining programming as the process of creating a sequence of instructions to automate tasks. We'll explore Computational Thinking, breaking complex problems into manageable steps, and the critical distinction between an **Algorithm** (the step-by-step procedure) and a **Program** (the implementation of the algorithm in code). Environment setup is crucial: Installation of **Anaconda** (a distribution for Python and R that simplifies package management), selection of an Integrated Development Environment (IDE) like **VS Code** or a cloud-based solution like **Google Colab**, and executing basic interactive commands within a **Jupyter Notebook** environment.

Day 2 — Variables & Data Types

Dive into the mechanics of Python. Understanding the importance of correct **Python Syntax** and the rules for creating **Identifiers** (names for variables, functions, etc.). Introduction to the essential print() function for output. Focus on Python's dynamic typing with core **Data Types**: int (integers), float (floating-point numbers), string (text), and boolean (True/False). Learn to inspect data types using the type() function, capture user input with input(), and master **type casting** (explicitly converting one data type to another).

Day 3 — String Manipulation & Formatting

Text data is ubiquitous. This day covers advanced handling of the string data type. Techniques include **String Indexing** (accessing individual characters), **Slicing** (extracting substrings), and essential **String Operations/Methods** such like .upper(), .lower(), .replace(), and .split(). Crucially, we master modern string formatting using **f-strings** (formatted string literals) for readable and efficient code output.

Day 4 — Data Structures & Mutability

A deep dive into Python's powerful built-in **Data Structures**. We explore **Lists** (ordered, mutable, allows duplicates), **Sets** (unordered, mutable, unique elements), **Tuples** (ordered, immutable, allows duplicates), and **Dictionaries** (unordered, mutable, key-value pairs). A key conceptual focus is understanding **Mutability vs Immutability**, which dictates how an object can be changed after creation and affects performance and safety.

Day 5 — Operators in Python

Mastering the tools for computation and logic. Comprehensive coverage of **Arithmetic Operators** (+, -, *, /, //, %, **), **Comparison Operators** (==, !=, >, <, >=, <=), and **Logical Operators** (and, or, not). Practical application of **Assignment Operators** (=, +=, -=, etc.) and **Membership Operators** (in, not in) for checking existence within sequences.

WEEK 2 · Control Flow, Functions, and I/O

Day 6 — Control Flow & Conditional Statements

Learning to control the execution path of a program. Implementation of the core decision structures: **if**, **elif** (else if), and **else** statements. Exploration of **nested conditions** for complex logic and applying conditional logic to build practical, real-world decision-making structures. Introduction to the range() function as preparation for loops.

Day 7 — Loops & Iterations

Automating repetitive tasks. Utilizing **while loops** (for indefinite iteration) and **for loops** (for definite iteration, typically over a sequence). Understanding **nested loops** for multi-dimensional operations (like matrices). Control loop behavior with **pass** (placeholder), **break** (exit loop), and **continue** (skip current iteration). Conclude with an introduction to efficient, concise code writing using **List and Dictionary comprehensions**.

Day 8

Functions Basics

The cornerstone of modular programming. Focus on **defining functions** (def), understanding the different **types of functions** (built-in, user-defined), and the critical concepts of **parameters** (inputs) and **return values** (outputs). Introduces the concept of **code optimization** and conciseness using single-expression **Lambda functions** (anonymous functions).

Day 9

Advanced Functions & Modules

Extending functional programming. Deep dive into **Higher-order functions**: **map** (apply a function to all items), **filter** (filter items based on a function), and **reduce** (cumulative computation). Learn to import and utilize external **modules** and packages, understanding **namespace** and **scope**. Utility functions like help() and dir() are introduced. Practical work with built-in modules: math, random, datetime, and os (Operating System).

Day 10

File Handling & Exceptions

Making programs robust and capable of data persistence. Detailed session on **handling errors and exceptions** using try, except, else, and finally blocks to gracefully prevent program crashes. Practical file input/output operations: **creating, reading, and writing files** in various common formats such as plain **TXT**, structured **CSV**, and semi-structured **JSON**.

WEEK 3 · Object-Oriented Programming (OOP), Regex, and Web Scraping

Day 11

Object-Oriented Programming (OOP) Fundamentals

Shifting to an object-centric paradigm. Understanding the **OOP Philosophy** and its advantages. Learning to create **Classes** (blueprints) and instantiate **Objects** (instances). Key components are **Attributes** (data/state) – specifically **Instance** and **Class Variables** – and the mandatory **__init__ constructor method** for object initialization.

Day 12

Advanced OOP Concepts

Deeper exploration of class functionality. Implementing **Instance Methods**, **Class Methods** (@classmethod decorator for methods bound to the class), and **Static Methods** (@staticmethod for utility functions within a class). Comprehensive coverage of the **Four Pillars of OOP**: **Encapsulation** (data hiding using access modifiers), **Inheritance** (creating new classes from existing ones), **Abstraction** (showing only essential information), and **Polymorphism** (one interface, many forms).

Day 13

OOP Implementation & Magic Methods

Practical application of OOP. Techniques include **Method Overriding** (redefining a parent's method in a child class) and using the **super() function** to call parent methods. Introduction to **Special/Magic Methods** (e.g., __str__, __len__). Concluding with the use of **Abstract Base Classes (ABC module)** for enforcing structure and a hands-on project: building a simple **Console-Based Student Management System** using all OOP concepts learned.

Day 14

Unstructured Data & Regular Expressions

Handling real-world data complexity. Differentiating between **Structured, Semi-Structured, and Unstructured Data**. Mastering **Regular Expressions (Regex)**, including **Regex Literals** and **Meta Characters** (., *, +, ?, [], etc.) for pattern matching. Practical focus on **cleaning text data** (e.g., removing noise) and validation tasks, such as **Email validation**.

Day 15

Web Scraping Basics

The first step in data acquisition from the web. Learn the concept of **Data Collection (web-scraping)**: programmatically extracting data from websites. The goal is converting the initial **unstructured web data (HTML)** into **structured formats** (like a Pandas DataFrame) suitable for analysis and generating insights.

WEEK 4 · NumPy, Pandas, and EDA Capstone

Day 16

NumPy (Numerical Python)

The backbone of numerical computing in Python. Introduction to the ndarray (N-dimensional array). Mastering **Array creation**, basic **printing**, **indexing**, **slicing**, and **reshaping** arrays. Key topics include fundamental **Basic operations**, **mathematical functions** (trigonometric, exponential), and the efficiency of **Broadcasting** and **Vectorization** when compared to traditional Python loops.

Day 17

Pandas Fundamentals

The essential library for data manipulation and analysis. Introduction to the two core data structures: **Series** (1-dimensional) and **DataFrames** (2-dimensional, tabular). Practical application in **Data importing and exporting** using common formats like **CSV**, **Excel**, **JSON files**, and even data extracted from **HTML** tables. Initial steps involve using basic data understanding operations like .head(), .info(), and .describe().

Day 18

Advanced Pandas & Data Manipulation

Deepening Pandas proficiency. Techniques for data selection: **Indexing** (.loc, .iloc), **Slicing**, and **Conditional Filtering**. Essential data summarization with **GroupBy operations**. Comprehensive coverage of **Handling Missing Data** (categorizing as MCAR, MAR, MNAR, and implementing Imputation techniques). Other cleaning tasks include **Removing Duplicates** and practical **String/DateTime Manipulations**.

Day 19

Data Merging & Feature Engineering

Integrating and preparing data for modeling. Techniques for combining DataFrames: **Concatenating** (stacking), **Merging** (SQL-style joins), and **Joining** (index-based joins). Introduction to **Window Functions** (e.g., rolling means). Essential **Feature Engineering** concepts: **Outlier detection** using statistical methods (**IQR, Z-score**) and a critical discussion on **addressing data leakage** to ensure model integrity.

Day 20

Month 1 Capstone & Consolidation

A culmination of Month 1 skills. This day focuses on integrated, **Comprehensive Case Studies** that require synthesizing knowledge of Python, functions, and data manipulation with Pandas. A key component is applying **Web Scraping from scratch** to acquire data, followed by thorough data cleaning and preparation using the advanced Pandas techniques learned.

Month 02 — Statistics, Exploratory Data Analysis & Probability

Weeks 5–8 · Days 21–40

WEEK 5 · Visualization and Descriptive Statistics

Day 21

Data Visualization with Matplotlib

The foundational plotting library. Understanding the **Principles of basic plotting** (figure and axes). Mastering the creation of **subplots** for multi-panel visualizations and customizing **properties** (titles, labels, colors). Hands-on building of fundamental plot types: **Line plots**, **Pie Charts**, **Bar Graphs**, **Histograms** (for distribution), and **Scatterplots** (for correlation).

Day 22

Data Visualization with Seaborn

A high-level, statistically-focused visualization library built on Matplotlib. Constructing complex statistical plots like **Box and Violin Plots** (for distributions and comparisons), **Joint Plots** (bivariate analysis), **Cat Plots** (categorical data), **Heatmaps** (for correlation matrices), **Pair Plots** (all pairwise relationships), and **Count plots** (frequency).

Day 23

Exploratory Data Analysis (EDA)

The art of uncovering insights before modeling. Understanding the **goals of EDA** (data cleaning, structure identification, outlier detection, hypothesis generation). Practical steps in **Uni-Variate** (single variable) and **Bi-Variate** (two variables) Analysis. Focusing on **Distribution analysis**, creating a **Correlation matrix** to quantify relationships, and **Trend identification** in time-series or sequential data.

Day 24

Descriptive Statistics

Quantifying data characteristics. Defining the distinction between **Population vs Sample** and **Parameter vs Statistic**. Reviewing **Data Types** (Nominal, Ordinal, Interval, Ratio). In-depth study of **Measures of Central Tendency**: **Mean** (average), **Median** (middle value), and **Mode** (most frequent value), and when to use each.

Day 25

Measures of Dispersion & Shape

Assessing data spread and form. Methods for quantifying variability: Calculating **Variance**, **Standard deviation** (the square root of variance), **Range**, and the **Interquartile Range (IQR)**. Understanding data shape using **Skewness** (asymmetry) and **Kurtosis** (peakedness). Utilizing **Box Plots** as a visual tool for measuring dispersion and effective **outlier detection**.

WEEK 6 · Probability and Distributions

Day 26

Probability Basics

The foundation of statistical inference. Defining **Experiments**, **Sample Space**, **Events**, and the calculation of classical probability. Covering the fundamental **Probability rules/axioms** (e.g., addition rule, multiplication rule).

Day 27

Conditional Probability

Probability based on prior information. Understanding the relationship between **Independent vs dependent events**. In-depth study and application of **Bayes Theorem** (reversing conditional probability) for practical scenarios like spam detection.

Day 28

Probability Distributions & Random Variables

Modeling uncertainty. Defining **Random Variables** (variables whose values are outcomes of random events). Distinguishing between **Discrete** (countable outcomes) and **Continuous** (any value in a range) probability distributions, and the difference between **PMF** (Probability Mass Function) and **PDF** (Probability Density Function).

Day 29

Advanced Distributions

Key distributions used in data science. Detailed analysis of **Bernoulli** (single trial), **Binomial** (multiple Bernoulli trials), and **Poisson** (count of events over time/space) distributions for discrete data. Mastering the **Normal Distribution** (bell curve), **Standard Normal Distribution** (Z-scores), **Z-distribution**, and **T-distribution** for continuous data and inference.

Day 30

Data Gathering & Sampling Methods

Techniques for collecting representative data. Reviewing different methods to select a subset of the population: **Convenience**, **Simple Random**, **Stratified** (proportional representation), **Cluster** (group-based), and **Systematic Sampling**.

WEEK 7 · Inferential Statistics and ML Introduction

Day 31

Inferential Statistics

Drawing conclusions about a population from a sample. Understanding **Sampling variability** and the cornerstone theorem: the **Central Limit Theorem** (CLT). Practical application in defining and calculating **Confidence Intervals** (CI) to estimate population parameters.

Day 32

Hypothesis Testing Fundamentals

Formalizing statistical decision-making. The process of **Setting Null (H_0) & alternative (H_a) hypotheses**. Understanding the critical concepts of **p-value**, **significance level (α)**, and the risks involved: **Type I Error** (α, false positive) and **Type II Error** (β, false negative).

Day 33

Parametric vs Non-Parametric Tests

Selecting the appropriate statistical test. Distinguishing between tests that assume a distribution (parametric) and those that do not (non-parametric). Executing tests for **normality** (e.g., Shapiro-Wilk), and implementing the workhorse tests: **Z-tests** (large sample, known σ) and **T-tests** (small sample, unknown σ).

Day 34

Advanced Hypothesis Testing

More complex comparative and association tests. Practical application of **A/B testing** (comparing two groups), **Chi-Square Tests** (for categorical association), **F-Tests** (comparing variances), and **ANOVA** (Analysis of Variance for comparing means of 3 or more groups).

Day 35

Introduction to Machine Learning Workflow

The transition to predictive modeling. Defining **Supervised vs Unsupervised Learning**. Overview of various **algorithms' approaches** (e.g., geometric, probabilistic). Critical discussion of the **Bias-variance tradeoff** as the central challenge in model building (underfitting vs. overfitting).

WEEK 8 · Pre-processing and Simple Regression

Day 36

Data Pre-processing for ML

Preparing data for optimal model performance. Techniques for pre-processing **Tabular, text, and image data**. Key focus on identifying and diagnosing model fit: **under fit** (high bias), **optimal fit**, and **over fit** (high variance) scenarios. Practical implementation of building robust data processing **pipelines using sklearn**.

Day 37

Model Validation Methods

Ensuring generalizability of the model. Implementing and understanding various methods to estimate a model's performance on unseen data: **Hold-out validation**, **Cross-Validation**, the exhaustive **Leave-One-Out Cross-Validation (LOOCV)**, and the industry-standard **k-Fold Cross-Validation**.

Day 38

Simple Linear Regression

The foundation of predictive modeling. Understanding the **Simple Linear Regression equation** ($\hat{y} = \beta_0 + \beta_1 x_1$). Techniques for **estimating coefficients** using the method of **Ordinary Least Squares (OLS)**. Crucial review of the **OLS Assumptions** (linearity, homoscedasticity, independence, normality of errors).

Day 39

Multiple & Polynomial Regression

Extending the linear model. Expanding to **Multiple Regression** ($\hat{y} = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n$). Techniques for **estimating regression coefficients** in multivariate settings. Identifying and mitigating **Multicollinearity**. Handling non-linear relationships with **Polynomial Regression**, including detecting **feature interactions**, and methods for **feature selection** (forward, backward, stepwise).

Day 40

Optimization & Regularization

Improving model training and preventing overfitting. Introduction to the fundamental optimization algorithm: **Gradient Descent** (including Batch, Stochastic, and Mini-Batch variants). Defining **Cost functions** (or Loss functions). Implementing **Regularization** techniques to penalize complex models: **L1 (Lasso)** (for feature selection), **L2 (Ridge)** (for shrinking coefficients), and **ElasticNet** (a combination of both).

Month 03 — Supervised & Unsupervised Machine Learning

Weeks 9–12 · Days 41–60

WEEK 9 · Regression and Classification Metrics

Day 41

Evaluation Metrics for Regression

Quantifying regression model performance. Deep dive into **Residual Analysis** (plotting errors). Using the **Q-Q Plot** to check for normality of residuals. Identifying the **line of best fit**. Essential metrics: **Mean Squared Error (MSE)**, **Root Mean Squared Error (RMSE)**, **R Squared** (coefficient of determination), and the penalty-adjusted **Adjusted R Squared**.

Day 42

Logistic Regression

Classification using a linear approach. Understanding the **Difference between Regression and Classification** (predicting a value vs a class). The mathematical basis: the **Logit/Sigmoid functions** to squash output between 0 and 1. Defining **decision boundaries** and the crucial process of **setting thresholds** to convert probabilities into discrete classes.

Day 43

Classification Evaluation Metrics

Comprehensive assessment of classification models. Mastering the **Confusion Matrix** (True Positives, False Negatives, etc.). Calculating key metrics: **Accuracy**, **Error rate**, **Precision** (purity), **Recall** (completeness), and the harmonic mean **F1 Score**. Understanding **TPR/FPR** (True/False Positive Rates) and the **ROC-AUC** (Receiver Operating Characteristic - Area Under the Curve) for threshold-independent performance.

Day 44

Naive Bayes Classifier

A fast and simple probabilistic classifier. Understanding the **Principles of Naive Bayes**, which assumes feature independence. Defining **Posterior** (what we want to find), **Prior** (initial belief), and **Likelihood** (how well data fits hypothesis) probability. Exploring different practical **types** of Naive Bayes: **Multinomial** (for counts/text), **Bernoulli** (binary features), **Gaussian** (continuous data), and **Categorical**.

Day 45

K-Nearest Neighbors (KNN)

A non-parametric, instance-based learning algorithm. Detailed exploration of the **K-Nearest Neighbor Algorithm** mechanics. Choosing appropriate **distance metrics** (Euclidean, Manhattan). The problem of the **Curse of Dimensionality** in KNN. Distinguishing between **Eager vs Lazy learners**. Practical application of **Hyperparameter tuning** (selecting the optimal 'K').

WEEK 10 · SVM, Trees, and Ensemble Methods

Day 46

Support Vector Machines (SVM) – Part 1

A powerful classifier based on hyperplanes. Understanding the core concept: the **Maximal Margin Classifier**. Defining **HyperPlanes**, **Support Vector Classifiers**, and the difference between **Hard vs Soft Margin Classification** (allowing some misclassification).

Day 47

Support Vector Machines (SVM) – Part 2

Handling non-linear data with SVM. Techniques for **Classification with Non-linear Decision Boundaries**. The genius of the **Kernel Trick** for mapping data to higher dimensions, focusing on the **Polynomial** and **Radial Basis Function (RBF)** kernels. Practical knowledge on **Tuning Hyper parameters** like Gamma, Cost (C), and Epsilon (ε).

Day 48

Decision Trees

An interpretable, rule-based model. Mastering **Basic Terminology**: **Root Node**, **Internal Nodes**, and **Terminal Node (Leaf)**. Distinguishing between **Classification vs Regression trees**. The process of evaluating **split quality** using metrics like the **Gini Index** and **Entropy** (information gain).

Day 49

Tree Optimization

Improving the performance and generalizability of trees. Methods for **Handling Overfitting** in complex trees through **Pruning** (reducing tree complexity) and setting **Stopping Criteria** (e.g., minimum leaf size). **Accuracy Estimation** using cross-validation. Implementing automated **Hyper parameter tuning** using efficient methods like **random search** and **grid search**.

Day 50

Ensemble Methods & Random Forest

Combining multiple models for better performance. Core **Concepts of Ensemble Learning** (bagging, boosting). Understanding **Bootstrap Aggregation (Bagging)**. Deep dive into **Random Forest mechanics**: utilizing random subsets of features and data. Determining **Variable selection** and **feature importance**. Analyzing the **OOB (Out-of-Bag) error** estimate.

WEEK 11 · Boosting and Dimensionality Reduction

Day 51

Boosting Algorithms

Sequentially building models to correct previous errors. Implementing classic and modern boosting techniques: **Adaboost** (Adaptive Boosting), **Gradient Boosting**, and the highly efficient **XG Boosting** (eXtreme Gradient Boosting). Comprehensive review of key **Hyper parameters**, **Pros**, and **Cons** of these powerful boosting methods.

Day 52 **Intro to Unsupervised Learning**

Discovering patterns in unlabeled data. Understanding the core **Challenges of Unsupervised Learning**. The key difference in goal: **Clustering** (grouping data) vs **Classification** (predicting a label).

Day 53 **Dimensionality Reduction – PCA**

Principal Component Analysis for feature space compression. Defining **Principal Components** (new orthogonal axes), **EigenValues** (variance along axes), and **EigenVectors** (direction of axes). The property of **Orthogonality**. The mathematical process of **Transforming data** and **calculating the proportion of variance explained** (scree plot analysis).

Day 54 **Dimensionality Reduction – t-SNE**

A non-linear technique for visualization. Understanding the complex **Mechanics of t-Distributed Stochastic Neighbor Embedding (t-SNE)** for visualizing high-dimensional data in 2D/3D space. **Comparative case studies** highlighting the strengths and weaknesses of **PCA** (linear) and **t-SNE** (non-linear, locally focused).

Day 55 **K-Means Clustering**

A centroid-based partitioning algorithm. The iterative process of **Establishing Centroids** and **Medoids**. Crucially, methods for **deciding the optimal 'k'** (number of clusters) using the **Elbow Method/Inertia** and evaluating cluster quality with the **Silhouette score**.

WEEK 12 · Advanced Clustering and Capstone

Day 56 **Hierarchical Clustering**

Building a hierarchy of clusters. Distinguishing between **Divisive** (top-down) vs **Agglomerative** (bottom-up) Clustering. Interpreting the results using **Dendrograms** (tree-like diagrams). Understanding different **Linkage Methods** (single, complete, average) and exploring **practical clustering applications**.

Day 57 **DBSCAN**

Density-Based Spatial Clustering of Applications with Noise. The core concepts of **Density-based clustering**. Identifying data points as **core points**, **border points**, and **noise** (outliers) based on neighborhood density.

Day 58 **Applied Machine Learning – Part 1**

Real-world application of ML models. Focus on advanced techniques for **Missing Value imputation** using sophisticated **Machine Learning Algorithms** (e.g., using regression to predict missing values, or MICE).

Day 59 **Applied Machine Learning – Part 2**

Another critical real-world application. Utilizing **Machine Learning Algorithms** (e.g., Isolation Forest, One-Class SVM) for **Outlier and Anomalies detection** in datasets, crucial for fraud detection or system monitoring.

Day 60 **Month 3 Capstone & ML Deployment Basics**

The final project for supervised/unsupervised learning and the first taste of MLOps. Developing and serving a functional **Machine Learning Application**. Utilizing **Streamlit** (a Python library) for creating a simple web interface. The culmination is a **Customer Segmentation** (unsupervised) or a **classification** (supervised) mini-project.

Month 04 — Deep Learning & Computer Vision

Weeks 13–16 · Days 61–80

WEEK 13 · Deep Learning Fundamentals and ANNs

Day 61 **Introduction to Deep Learning**

Setting the context for neural networks. Clarifying the hierarchy: **AI vs ML vs DL**. A brief **History of Deep Learning** (from perceptrons to modern ANNs). Understanding the functional analogy between **Biological Neurons vs Perceptrons** (the fundamental building block).

Day 62

Deep Learning Frameworks

Introducing the tools of the trade. Setting up and utilizing **Google Colab** for cloud-based GPU access. Introducing **TensorFlow** (Google's framework) with **basic syntax** for tensor operations. Understanding **TensorFlow Graphs** and using **TensorBoard** for model visualization and debugging.

Day 63

Artificial Neural Networks (ANN)

The core structure of Deep Learning. Understanding the **Multi-Layer Perceptron (MLP) Architecture** (input, hidden, output layers). The process of **Forward Propagation** (data flowing through the network) and training a simple **single neuron model** (perceptron).

Day 64

Backpropagation Mechanics

The engine of neural network learning. In-depth understanding of the **chain rule** in calculus as applied to network training. The process of **memorization in Backpropagation** (adjusting weights based on error) and the critical step of **calculating weight updates** for each layer using the computed gradients.

Day 65

Activation Functions

Introducing non-linearity into the network. Addressing the critical issues of the **vanishing and exploding gradient problems**. Implementing and understanding the properties of classic and modern functions: **Sigmoid**, **Tanh**, **ReLU** (Rectified Linear Unit), and the **ReLU Variants** (**Leaky, Parametric, ELU, SELU**).

WEEK 14 · Optimization, Keras, and Image Prep

Day 66

Optimizers & Loss Functions

Driving the network towards convergence. Distinguishing between **Convex vs Non-Convex Functions** in the loss landscape. Identifying and navigating **Saddle Points**. Implementing and comparing various advanced **Optimizers**: **SGD with Momentum, NAG, RMSprop, AdaDelta, AdaGrad, ADAM**, and **NADAM**.

Day 67

Regularization & Tuning in DL

Strategies for building robust, non-overfitting networks. Overcoming **DNN failures** by implementing **Dropouts** (randomly ignoring neurons), **Batch Normalization** (stabilizing learning), various **Weight Initialization Techniques** (e.g., Xavier, He), and **Early Stopping** (halting training when validation loss stalls).

Day 68

Applied Keras & TensorFlow (Regression)

Hands-on model construction. Focus on the high-level API **Keras** for simplified network building. **Hands-on model building**: Constructing a **Regression model** (predicting a continuous value) using the **MLP** architecture with the **Keras Sequential API**.

Day 69

Deep Learning for Classification

Applying ANNs to classification tasks. **Hands-on model building**: Constructing a **Classification model** (predicting a category) using the **MLP** architecture. Implementing appropriate activation functions (e.g., softmax) and loss functions (e.g., cross-entropy) in **Keras**.

Day 70

Image Processing Foundations

The gateway to Computer Vision. Introduction to **OpenCV** (Open Source Computer Vision Library). Understanding **how images are formed/stored** (pixels, channels). Exploring different **color spaces** (RGB, Grayscale). Practical operations: **reading/writing/saving images**, and capturing live data **via webcam**.

WEEK 15 · Convolutional Neural Networks (CNN)

Day 71

Advanced OpenCV

Extending image manipulation. Techniques for **Drawing on images/videos** (shapes, text). Understanding **bitwise operators** for image masking. Building **image histograms** to analyze intensity distribution. Distinguishing between **Affine vs Non-Affine Transformations** for geometric adjustments.

Day 72 **Convolutional Neural Networks (CNN) Intro**

The architecture for image understanding. The fundamental question: **Why CNN over MLP for images?** Detailed breakdown of **how convolution works** (filters/kernels), the purpose of **Padding** (maintaining size), **Stride** (movement of the filter), and the function of **Pooling layers** (downsampling).

Day 73 **CNN Architectures – Part 1**

Studying landmark network designs. Understanding and implementing the structure of **early network structures**: **LeNet5** (pioneering CNN for digit recognition) and **AlexNet** (the winner of the 2012 ImageNet challenge, proving the power of deep CNNs).

Day 74 **CNN Architectures – Part 2**

Deepening the understanding of state-of-the-art models. In-depth study of the architectures: **VGG16** and **VGG19** (focus on depth using 3x3 kernels), **Inception Net (GoogLeNet)** (using multi-size filters), and **ResNet (Residual Networks)** (solving the degradation problem with skip connections).

Day 75 **Advanced Architectures & Case Study**

Exploring highly efficient and modern designs. Introduction to **Xception** (depthwise separable convolution), **MobileNet** (efficient for mobile/edge devices), and **EfficientNet** (scaled architecture). Concluding with an **Image Classification Case Study: Face Mask Detection**.

WEEK 16 · Transfer Learning and Object Detection

Day 76 **Transfer Learning**

Leveraging pre-trained knowledge. Defining **What is Transfer Learning?** Understanding **When and why to use it** (e.g., small datasets). Practical skill: **Coding Transfer Learning pipelines** (feature extraction and fine-tuning) for **custom datasets** (e.g., Bone fracture multi-region detection).

Day 77 **CNN Hyperparameter Tuning**

Optimizing Deep Learning models. Utilizing automated search techniques for model optimization: **Random Search**, the highly efficient **Hyperband**, and the probabilistic **Bayesian optimization**.

Day 78 **Object Detection Fundamentals**

Locating and classifying multiple objects in an image. Introduction to the classic **Object Detection algorithms**: **R-CNN** (Region-based CNN), **Fast R-CNN**, and the region proposal network-based **Faster R-CNN**.

Day 79 **YOLO (You Only Look Once)**

The modern, real-time object detection standard. Understanding the **YOLO algorithm mechanics** (a single-shot detector). Practical **Implementation of YOLO v7/v8 using Ultralytics** for high-speed, accurate detection.

Day 80 **Month 4 Capstone**

The final Computer Vision project. A complete, integrated project: **Building a real-time Helmet Detection or Plant Disease Prediction system** using the latest techniques like **YOLO** for detection/localization and **Transfer Learning** for backbone model utilization.

Month 05 — Natural Language Processing (NLP) & Generative AI

Weeks 17–20 · Days 81–100

WEEK 17 · NLP Foundations and Word Embeddings

Day 81 **Introduction to NLP & Text Preprocessing**

The start of language understanding. Overview of NLP applications. Core toolkits: **NLTK** (Natural Language Toolkit) and **spaCy** (industrial-strength NLP). Essential cleaning steps: **Tokenization** (breaking text into words/sentences), **removing special characters**, and **Stop words** removal (common words like 'the', 'a').

Day 82

Stemming, Lemmatization & Vectorization

Standardizing and quantifying text. Differences between **Stemming** (crude removal of suffixes) and **Lemmatization** (linguistic root form). Techniques for converting text to numerical vectors: **Bag of Words (BOW)** (count-based) and **TF-IDF** (Term Frequency-Inverse Document Frequency, importance-based) implementation using NLTK/sklearn.

Day 83

Word Embeddings

Capturing semantic context numerically. Moving beyond sparse vectors. Understanding the core concept of **Word2Vec** (predicting context or word): **Skip-Gram** (predicting context from word) & **CBOW** (Continuous Bag of Words - predicting word from context). Introduction to other dense vector models: **GloVe** and **FastText**.

Day 84

Sequential Data & RNNs

Handling ordered data like sentences. Defining the need: **Why use Recurrent Neural Networks (RNNs)?** Understanding **How RNNs work** (hidden state maintaining memory), the process of **training RNNs**, and the critical challenge of **handling the vanishing gradient** in long sequences.

Day 85

Advanced RNNs & Sequence Tagging

Improving memory and applying to specific tasks. Implementing **Bi-Directional RNNs** (processing sequence forward and backward). Practical **Case Study: Named Entity Recognition (NER)** (identifying entities like people/places) and **POS Tagging** (Part-of-Speech tagging).

WEEK 18 · Advanced Sequence Models and Transformers

Day 86

LSTM & GRU

The solution to the vanishing gradient problem. Detailed study of the **architecture of Long Short-Term Memory (LSTM)** networks and the simplified **Gated Recurrent Units (GRU)**. Understanding the function of the internal **gates** (input, forget, output) and **memory cells** for long-term dependence.

Day 87

Autoencoders

Learning efficient data representations. Understanding the **Encoder-Decoder Architecture** (compression and reconstruction). Exploring different **Types of autoencoders** (denoising, variational). **Case study on image compression/reconstruction** using the MNIST dataset.

Day 88

Attention Mechanisms

The revolution in sequence-to-sequence models. A deep dive into **How Attention models revolutionized NLP**. The **Working principles of Attention** (selectively focusing on important parts of the input sequence).

Day 89

Transformers

The backbone of modern LLMs. The complete **Transformer architecture**: **Self-Attention** (contextualizing each token), **Multi-Head Attention** (multiple attention mechanisms in parallel), and the encoder/decoder stacks. Practical **Coding Transformers** for various text generation or translation tasks.

Day 90

BERT & Advanced NLP

Contextual word embeddings. Introduction to **BERT (Bidirectional Encoder Representations from Transformers)**, explaining **how it works** (masked language modeling, next sentence prediction), and the process of **fine-tuning BERT** for specific tasks. **Case Study: Text Classification and Sentiment Analysis** using a pre-trained BERT model.

WEEK 19 · Generative AI and Commercial APIs

Day 91

Introduction to Generative AI

The new paradigm of content creation. Defining **Generative vs Predictive AI**. Comprehensive **Overview of Large Language Models (LLMs)** (architecture, scale) and their vast **industry applications** (chatbots, coding, creative content).

Day 92 **Prompt Engineering**

The key skill for interacting with LLMs. Mastering strategic prompting techniques: **Zero-shot**, **One-shot**, and **Few-shot prompting**. Principles of **Instruction-based prompt design**, advanced **Chain-of-Thought techniques**, and output control using parameters like **temperature** and **tokens**.

Day 93 **Working with OpenAI**

Accessing industry-leading models. Steps for **Setting up and authenticating the OpenAI API**. Practical **Text generation** using powerful models like **GPT-3.5/GPT-4/GPT-4o**.

Day 94 **Multimodal GenAI**

Generating content across modalities. Understanding the architecture and capabilities of models like **DALL-E** and the process of **image generation**. **Hands-on project: Generating images from textual descriptions** using an API.

Day 95 **Gemini AI**

Google's state-of-the-art model. Steps for **Obtaining Gemini API keys**, exploring the different **Gemini models** (Nano, Pro, Ultra), and a **step-by-step creation of an AI-powered chatbot** using the Gemini API.

WEEK 20 · Open-Source LLMs and Deployment

Day 96 **Open-Source LLMs (LLaMA)**

Exploring community-driven models. Introduction to **Meta LLaMA** (Large Language Model Meta AI), its foundational **model architecture**, and exploring various **variants** (8B, 13B, 30B, 70B). Setting up and running the model in a **Local setup** and performing text generation.

Day 97 **LangChain Framework – Part 1**

Building complex LLM applications. Understanding the **Purpose and core components of LangChain** (Models, Prompts, Chains, Agents). **Setup, dependencies**, and **basic configuration** for building interconnected LLM workflows.

Day 98 **LangChain Framework – Part 2**

Implementing advanced LLM architectures. **Building real-world RAG (Retrieval-Augmented Generation) applications** with LangChain, which allows LLMs to access external, non-training data sources for grounding answers.

Day 99 **No-Code/Low-Code AI Agents & Ethics**

Autonomous systems and responsible development. Exploring the concept of **Autonomous agent risk assessment**. Critical discussions on model **bias**, **hallucination**, establishing **guardrails** for safe usage, and following **Responsible AI guidelines**.

Day 100 **Month 5 Capstone & Deployment**

The final Generative AI project and MLOps completion. Introduction to **Git basics** and **GitHub workflows** for version control. Best practices for **model saving** (serialization). The ultimate goal: **Deploying a Generative AI Writing Assistant/Chatbot** using **Streamlit** to create a live, accessible web application.

Month 06 — SQL Complete & Business Intelligence (Power BI)

Weeks 21–24 · Days 101–120

WEEK 21 · SQL Data Definition and Manipulation

Day 101 **Introduction to Databases**

The foundational concepts of data storage. Defining **What is Data**, the purpose of **DBMS vs RDBMS**, and the difference between **SQL vs MySQL vs NoSQL**. Understanding the **Client-Server Architecture** of databases and introduction to a client tool like **Workbench**.

Day 102

SQL Data Definition & Constraints

Structuring the database. Defining standard **Data types** (INT, VARCHAR, DATE). Enforcing data integrity using **Constraints** (**PRIMARY KEY, NOT NULL, UNIQUE, DEFAULT, CHECK, FOREIGN KEY**). Mastering **DDL Commands** (Data Definition Language): CREATE, ALTER, DROP, TRUNCATE.

Day 103

Data Manipulation & Querying

Adding, modifying, and retrieving data. Mastering **DML Commands** (Data Manipulation Language): INSERT, UPDATE, DELETE. The core of SQL: **DQL (Data Query Language)**: SELECT, limiting results with LIMIT, and finding unique values with DISTINCT.

Day 104

Data Filtering & Operators

Selecting specific records from a table. Filtering data using the essential **WHERE clause**. Advanced filtering with **Operators**: pattern matching with LIKE and **REGEXP** (regular expressions), range checking with **BETWEEN**, and matching a list of values with **IN**.

Day 105

Grouping & Sorting Data

Summarizing and ordering query results. Aggregating rows using **GROUP BY**. Filtering aggregated results with **HAVING**. Ordering the final output with **ORDER BY**. Using conditional logic within queries via **CASE statements**. Crucial understanding of the **order of query execution** (SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY).

WEEK 22 · Advanced SQL, Joins, and Windows

Day 106

Relational Databases & Keys

Modeling relationships between tables. Understanding the types of **Keys**: **Primary keys**, **Composite keys**, and **Foreign keys**. Defining **Types of relationships** (**1:1, 1:M, M:M**). Introduction to **ER diagrams** (Entity-Relationship diagrams).

Day 107

SQL Joins

Combining data from multiple tables. Implementing and understanding the differences between **Inner join** (intersection), **Left join** (all from left), **Right join** (all from right), **Outer join** (all rows), **Self join**, and **Cross join**.

Day 108

Set Operations & Subqueries

Combining query results and nesting queries. Using **Set Operations** to combine results with same columns: **UNION**, **UNION ALL**, **EXCEPT**, and **INTERSECT**. Writing and applying various **Subqueries**: **Scalar** (returns a single value), **Multiple** (returns a list of values), and **Correlated Subqueries** (dependent on outer query).

Day 109

Advanced Querying

Techniques for structured, complex queries. Using **Derived tables** (subqueries in the FROM clause). Mastering **Common Table Expressions (CTEs)** (named temporary result sets) for query organization. Introduction to **Recursive CTEs** for hierarchical data.

Day 110

Advanced SQL Objects

Beyond basic data retrieval. Deep dive into **Window Functions** for analytical tasks: **LEAD, LAG** (accessing rows above/below), **FIRST_VALUE, NTILE**. Understanding database objects: **Views** (stored queries), **Stored Procedures** (stored SQL code), **Triggers** (actions that fire automatically). Introduction to **TCL (Transaction Control Language)**: COMMIT, ROLLBACK.

WEEK 23 · Excel and Power BI Foundations

Day 111

Excel for Analytics Core

Reviewing essential Excel skills for data professionals. Review of **advanced Excel capabilities**: **Statistical/Logical functions** (e.g., IF, AND, OR), powerful conditional aggregations like **SUMIFS, COUNTIFS**, modern **Dynamic arrays**, lookup functions **VLOOKUP, HLOOKUP**, and practical **Date functions**.

Day 112

Introduction to Power BI

The start of Business Intelligence. Defining **What is BI?** Comprehensive **Power BI Desktop overview** (components). Understanding the **Power BI workflow** (Get Data → Transform → Model → Visualize). Reviewing **Data Import options** and navigating the **UI**.

Day 113

Power Query

The Extract, Transform, Load (ETL) layer. Introduction to **Power Query** for data transformation. Briefing on **M Language** (the Power Query language). Hands-on with essential transformations: **changing datatypes**, **filtering**, and various **inbuilt row/column transformations** (e.g., unpivot, pivot).

Day 114

Advanced Power Query

Complex data shaping techniques. Methods for combining datasets: **Appending** (stacking) and **Merging Queries** (joining). Creating utility columns using **Indexing** and **conditional columns** based on logic.

Day 115

Data Modeling (Power Pivot)

Structuring data for analysis. Introduction to **Data Modeling** (Star/Snowflake Schema). Defining **Relationship Cardinality** (**1:1, 1:M, M:M**) and understanding **cross-filter direction**. Practical use of the **Relationship views** to build a robust data model.

WEEK 24 · DAX, Visualization, and Final Capstone

Day 116

Introduction to DAX

Data Analysis Expressions for custom calculations. Key difference: **Calculated Columns vs Measures** (row vs filter context). Understanding **DAX syntax**, including **Logical functions**, **Text functions**, **Math**, and basic **Statistical functions**.

Day 117

Advanced DAX

Mastering context transition and time intelligence. Advanced **Aggregation functions** like **SUMX** and **AVERAGEX** (iterators). The most important function: the powerful **CALCULATE** (filter context modification). Understanding **Filter functions** like **ALL** and **ALLEXCEPT**. Implementing **Time Intelligence** using a **Date Dimension table**.

Day 118

Interactive Visualizations

Building compelling and dynamic reports. Creating standard visuals: **Categorical** (Bar/Column) and **Trend visuals** (Line/Area). Utilizing **Slicers**, performance indicators like **KPI cards**, and detailed **Matrices**. Advanced techniques: **Drill Through**, **Drilldown**, and integrating **Python Script Visuals**.

Day 119

Power BI Service & Security

Publishing and sharing the final report. The process of **Publishing reports** from Desktop to Service. **Dashboard creation/formatting** (pinning visuals). **Workspace sharing** for collaboration. Implementing **Row Level Security (RLS)** by defining roles and testing to ensure data access restrictions.

Day 120

Final Program Capstone

The grand finale: an end-to-end data science and business intelligence project. **End-to-End Business Analytics System**: **Extracting data using SQL**, **cleaning in Python/Pandas**, creating a **predictive model** (ML), and finally **serving the insights** via a deployed **Power BI Dashboard** with a **GenAI-generated executive summary** (connecting all six months of learning).

PROJECTS & CAPSTONES

13 things you'll build

DAY 13

Student Management System

A console application that puts classes, inheritance, magic methods and abstract base classes to work.

DAY 20

Web-Scraping Case Studies

Acquire data from the web from scratch, then clean and prepare it with advanced Pandas.

DAY 60

ML App with Streamlit

Customer segmentation (unsupervised) or classification (supervised), served as a live web app.

DAY 75

Face Mask Detection

An image-classification case study using modern, efficient CNN architectures.

DAY 76

Bone-Fracture Detection

Transfer-learning pipelines with feature extraction and fine-tuning on a custom dataset.

DAY 80

Real-Time Helmet Detection / Plant Disease

YOLO for detection and localisation with a transfer-learning backbone.

DAY 87

Autoencoder Reconstruction

Compress and reconstruct MNIST images with an encoder–decoder network.

DAY 90

BERT Sentiment Analysis

Fine-tune a pre-trained BERT model for text classification and sentiment.

DAY 94

Text-to-Image Generator

Generate images from textual descriptions with a multimodal GenAI API.

DAY 95

Gemini AI Chatbot

Step-by-step creation of an AI-powered chatbot using the Gemini API.

DAY 98

RAG Application

Retrieval-Augmented Generation with LangChain to ground answers in your own data.

DAY 100

GenAI Writing Assistant

A chatbot / writing assistant deployed live with Streamlit, version-controlled on GitHub.

DAY 120

End-to-End Analytics System

SQL extraction → Pandas cleaning → predictive model → Power BI dashboard with a GenAI executive summary.

Frequently asked questions

Do I need to know programming already?
No. Month 1 starts with what programming is, computational thinking and environment setup (Anaconda, VS Code, Google Colab, Jupyter), then builds up to OOP, NumPy and Pandas.

How much mathematics is involved?
Month 2 is dedicated to statistics and probability — descriptive statistics, distributions, sampling, confidence intervals and hypothesis testing — taught with practical data-science applications.

What deep-learning topics are covered?
Neural networks from perceptrons to backpropagation, activation functions, optimisers, regularisation, Keras and TensorFlow, OpenCV, CNN architectures from LeNet to EfficientNet, transfer learning, R-CNN and YOLO.

What will I do with Generative AI?
Prompt engineering, the OpenAI API, DALL·E image generation, Gemini chatbots, local LLaMA models, LangChain and RAG — ending with a GenAI writing assistant deployed on Streamlit.

Why are SQL and Power BI in a data-science program?
Real projects start in databases and end in dashboards. Month 6 covers SQL end to end and Power BI, culminating in a capstone that connects all six months.

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